Hotel Room Reservation System Project Report

Sri Lanka Institute of Information Technology



SE1020 – Object Oriented Programing

Year 1, Semester 2 - 2026

Hotel Room Reservation System

Project Report

WD38

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IT26101177	Jayarathne W.D.S.S
IT26100762	Randika N. G. R.
IT26100939	Punchihewa P.B.M.P.D.G
IT26100545	Rivisahan E.G.R

1. Project Overview

The Orchid Estate Hotel Room Reservation System is a web application that lets a boutique hotel manage its rooms, guests and bookings in one place. Guests register, browse and book rooms, manage their own bookings and review completed stays. Staff check guests in and record payments at the front desk, and administrators manage rooms, staff accounts, reviews and the home page content.

The system is built with Java 17 and Spring Boot, using JSP pages styled with Tailwind CSS for the user interface and a MySQL database accessed through Spring's JdbcTemplate. Every module follows the same layers: model classes, a DAO that runs the SQL, a service that holds the business rules and permission checks, and servlets that handle the web requests.

The project applies the core OOP principles throughout. A single abstract `User` class is the base for guests, staff and admins, and its permission methods deny by default and are overridden by each role. Rooms, reservations, payments and reviews each have their own inheritance hierarchy, where subclasses override behaviour such as pricing, cancellation rules, discounts and how a review is shown.

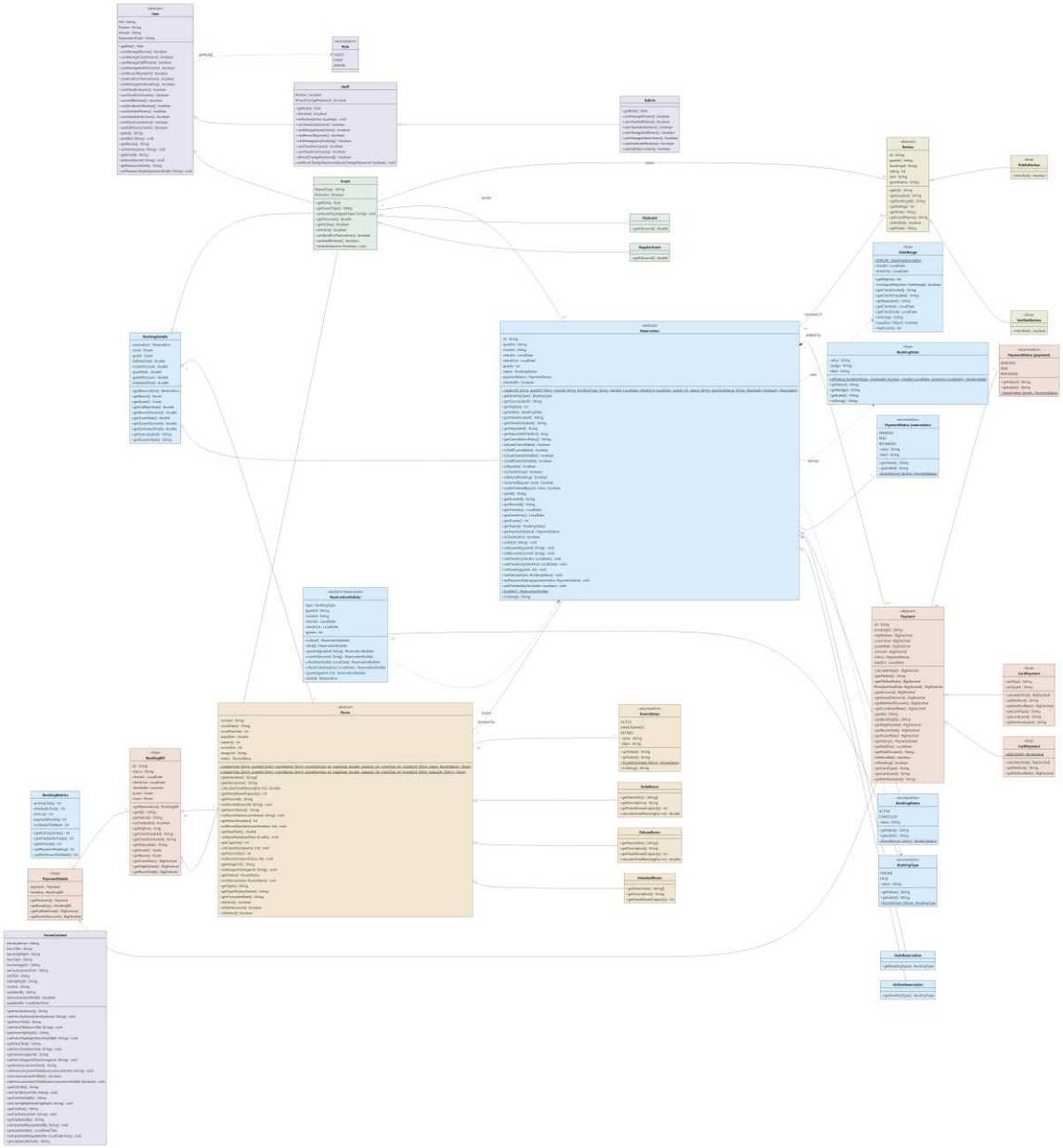
2. Workload Distribution

Each member owned one module end to end: its model classes, database table, service, servlets and pages.

IT Number	Module	OOP Concepts	CRUD Operations
IT26100545	Guest / User Management	Encapsulation, Inheritance, Polymorphism	Create: guest registers an account. Read: view and search guest profiles. Update: edit profile, staff grant VIP. Delete: soft-delete (deactivate) an account.
IT26100762	Room Management	Encapsulation, Inheritance, Polymorphism	Create: add a room. Read: search and filter rooms. Update: edit room details and status. Delete: soft-delete (retire) a room.
IT26100826	Reservation / Booking	Encapsulation, Abstraction, Polymorphism	Create: book a room. Read: view bookings and check availability. Update: change dates, check in. Delete: soft-delete (cancel) a booking.
IT26101177	Payment & Billing	Encapsulation, Inheritance, Polymorphism	Create: record a payment. Read: view invoices and payment history. Update: change status (pending, paid, refunded). Delete: delete a payment.
IT26100781	Admin / Staff Management	Encapsulation, Inheritance, Abstraction	Create: register a staff or admin account. Read: view staff accounts. Update: change roles and details. Delete: hard-delete a staff account.
IT26100939	Feedback / Review Management	Encapsulation, Inheritance, Polymorphism	Create: write a review. Read: view reviews. Update: edit own review, admins moderate. Delete: delete a review.

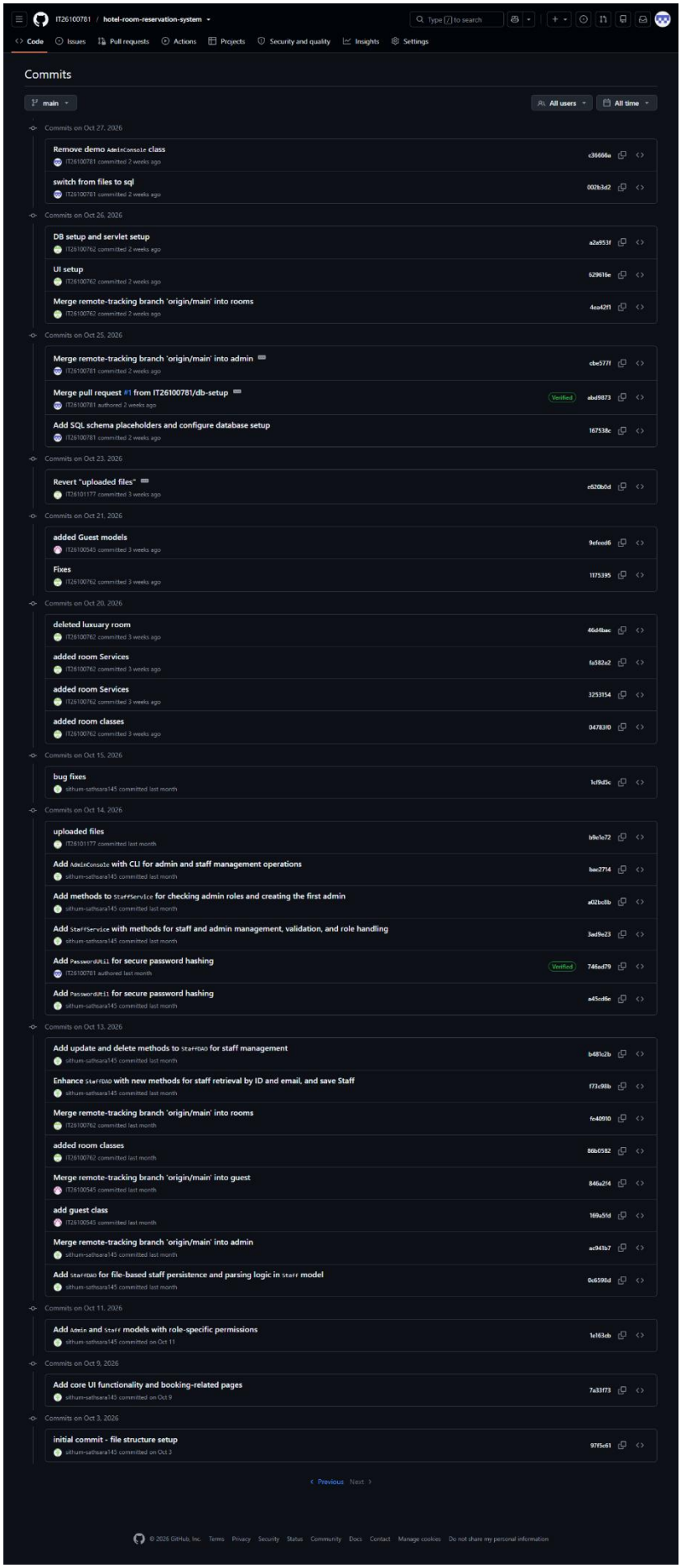
3. Domain Model Class Diagram

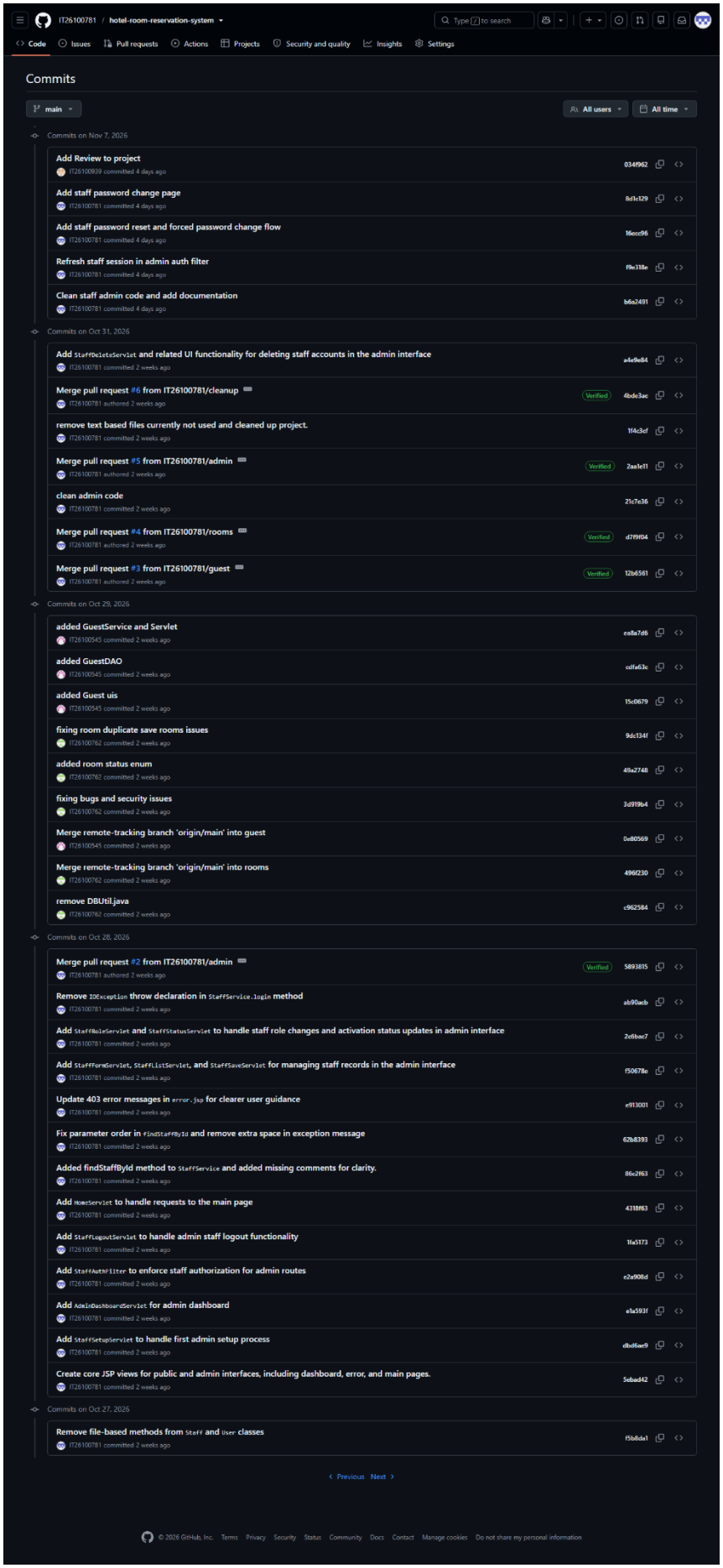
The diagram shows all 33 model classes with their attributes and methods, and only the relationships between models. DAOs, services and servlets are left out. Colours mark the module each class belongs to.

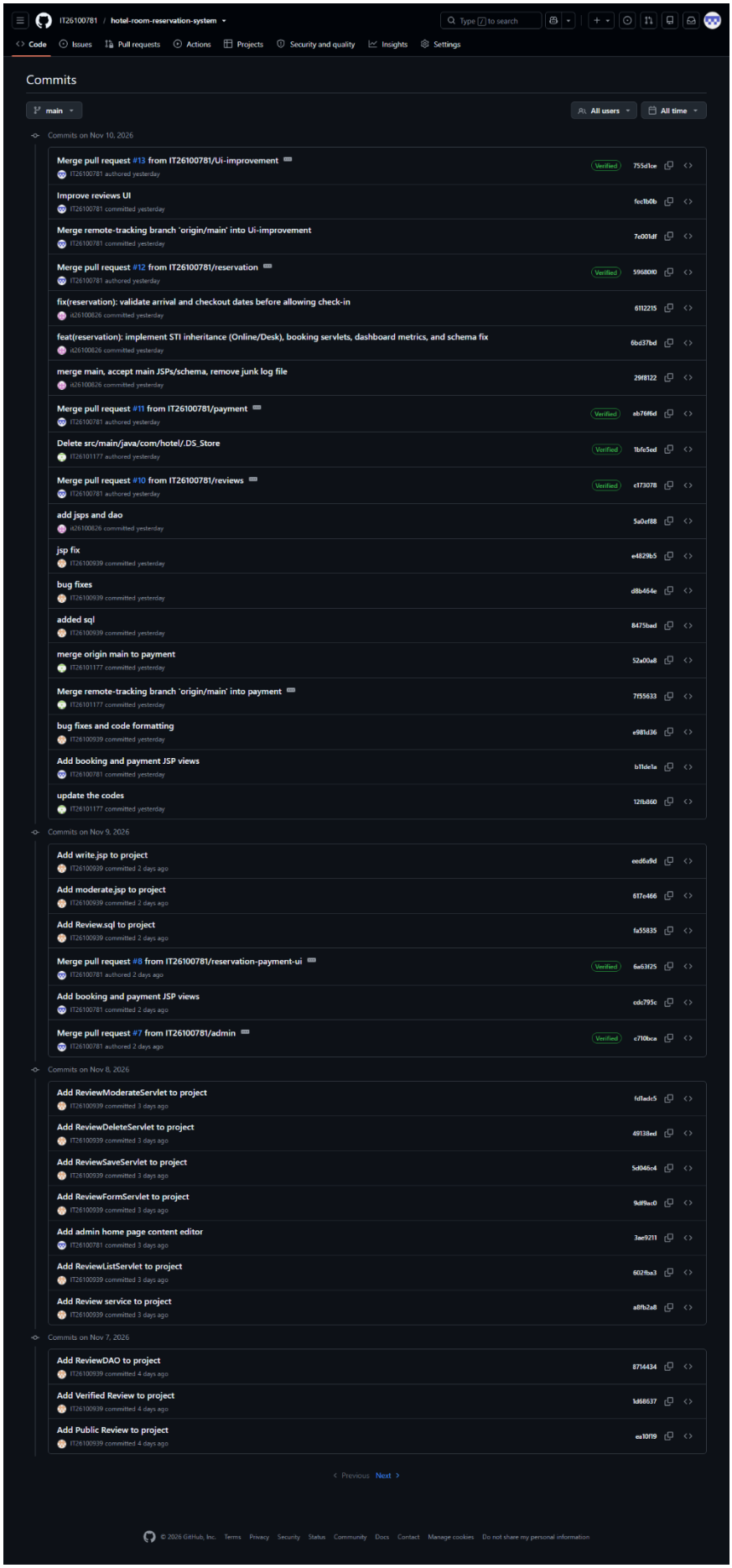


4. Commit History

GitHub commit history of the project repository.


IT26100781 / hotel-room-reservation-system
Type [/] to search
Code
Issues
Pull requests
Actions
Projects
Security and quality
Insights
Settings
Commits
main
All users
All time
Commits on Oct 27, 2026
Remove demo AdminConsole class
IT26100781 committed 2 weeks ago
c36666a
switch from files to sql
IT26100781 committed 2 weeks ago
002b3d2
Commits on Oct 26, 2026
DB setup and servlet setup
IT26100762 committed 2 weeks ago
a2a953f
UI setup
IT26100762 committed 2 weeks ago
629616e
Merge remote-tracking branch 'origin/main' into rooms
IT26100762 committed 2 weeks ago
4ea42f1
Commits on Oct 25, 2026
Merge remote-tracking branch 'origin/main' into admin
IT26100781 committed 2 weeks ago
cbe577f
Merge pull request #1 from IT26100781/db-setup
IT26100781 authored 2 weeks ago
Verified
abd8873
Add SQL schema placeholders and configure database setup
IT26100781 committed 2 weeks ago
167538c
Commits on Oct 23, 2026
Revert "uploaded files"
IT26101177 committed 3 weeks ago
e620b0d
Commits on Oct 21, 2026
added Guest models
IT26100543 committed 3 weeks ago
9efeed6
Fixes
IT26100762 committed 3 weeks ago
1175395
Commits on Oct 20, 2026
deleted luxuary room
IT26100762 committed 3 weeks ago
46d4bac
added room Services
IT26100762 committed 3 weeks ago
fa582e2
added room Services
IT26100762 committed 3 weeks ago
3253154
added room classes
IT26100762 committed 3 weeks ago
04783f0
Commits on Oct 15, 2026
bug fixes
sithum-sathsara145 committed last month
1cf9d5c
Commits on Oct 14, 2026
uploaded files
IT26101177 committed last month
b9e1e72
Add AdminConsole with CLI for admin and staff management operations
sithum-sathsara145 committed last month
bac2714
Add methods to StaffService for checking admin roles and creating the first admin
sithum-sathsara145 committed last month
a02bc8b
Add StaffService with methods for staff and admin management, validation, and role handling
sithum-sathsara145 committed last month
3ad9e23
Add PasswordUtil for secure password hashing
IT26100781 authored last month
Verified
746ad79
Add PasswordUtil for secure password hashing
sithum-sathsara145 committed last month
a45cd6e
Commits on Oct 13, 2026
Add update and delete methods to StaffDAO for staff management
sithum-sathsara145 committed last month
b481c2b
Enhance StaffDAO with new methods for staff retrieval by ID and email, and save Staff
sithum-sathsara145 committed last month
f73c98b
Merge remote-tracking branch 'origin/main' into rooms
IT26100762 committed last month
fe40910
added room classes
IT26100762 committed last month
86b0582
Merge remote-tracking branch 'origin/main' into guest
IT26100543 committed last month
846a2f4
add guest class
IT26100543 committed last month
169a5fd
Merge remote-tracking branch 'origin/main' into admin
sithum-sathsara145 committed last month
ac941b7
Add StaffDAO for file-based staff persistence and parsing logic in Staff model
sithum-sathsara145 committed last month
0c6598d
Commits on Oct 11, 2026
Add Admin and Staff models with role-specific permissions
sithum-sathsara145 committed on Oct 11
1e163cb
Commits on Oct 9, 2026
Add core UI functionality and booking-related pages
sithum-sathsara145 committed on Oct 9
7a33f73
Commits on Oct 3, 2026
initial commit - file structure setup
sithum-sathsara145 committed on Oct 3
97f5c61
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Commits
main
All users
All time
Commits on Nov 7, 2026
Add Review to project
IT26100939 committed 4 days ago
034f962
Add staff password change page
IT26100781 committed 4 days ago
8d1e129
Add staff password reset and forced password change flow
IT26100781 committed 4 days ago
16ecc96
Refresh staff session in admin auth filter
IT26100781 committed 4 days ago
f9e318e
Clean staff admin code and add documentation
IT26100781 committed 4 days ago
b6e2491
Commits on Oct 31, 2026
Add StaffDeleteServlet and related UI functionality for deleting staff accounts in the admin interface
IT26100781 committed 2 weeks ago
a4e9e84
Merge pull request #6 from IT26100781/cleanup
IT26100781 authored 2 weeks ago
Verified
4bde3ac
remove text based files currently not used and cleaned up project.
IT26100781 committed 2 weeks ago
11f4c3cf
Merge pull request #5 from IT26100781/admin
IT26100781 authored 2 weeks ago
Verified
2aa1e11
clean admin code
IT26100781 committed 2 weeks ago
21e7e36
Merge pull request #4 from IT26100781/rooms
IT26100781 authored 2 weeks ago
Verified
d7f9f04
Merge pull request #3 from IT26100781/guest
IT26100781 authored 2 weeks ago
Verified
12b6561
Commits on Oct 29, 2026
added GuestService and Servlet
IT26100545 committed 2 weeks ago
ea8a7d6
added GuestDAO
IT26100545 committed 2 weeks ago
cdfa63c
added Guest uis
IT26100545 committed 2 weeks ago
15c0679
fixing room duplicate save rooms issues
IT26100762 committed 2 weeks ago
9dc134f
added room status enum
IT26100762 committed 2 weeks ago
49a2748
fixing bugs and security issues
IT26100762 committed 2 weeks ago
3d919b4
Merge remote-tracking branch 'origin/main' into guest
IT26100545 committed 2 weeks ago
0e80569
Merge remote-tracking branch 'origin/main' into rooms
IT26100762 committed 2 weeks ago
496f230
remove DBUtil.java
IT26100762 committed 2 weeks ago
c962584
Commits on Oct 28, 2026
Merge pull request #2 from IT26100781/admin
IT26100781 authored 2 weeks ago
Verified
5893815
Remove IOException throw declaration in StaffService.login method
IT26100781 committed 2 weeks ago
ab90acb
Add StaffRoleServlet and StaffStatusServlet to handle staff role changes and activation status updates in admin interface
IT26100781 committed 2 weeks ago
2c6bac7
Add StaffFormServlet, StaffListServlet, and StaffSaveServlet for managing staff records in the admin interface
IT26100781 committed 2 weeks ago
f50678e
Update 403 error messages in error.jsp for clearer user guidance
IT26100781 committed 2 weeks ago
e913001
Fix parameter order in findStaffById and remove extra space in exception message
IT26100781 committed 2 weeks ago
62b8393
Added findStaffById method to StaffService and added missing comments for clarity.
IT26100781 committed 2 weeks ago
86e2f63
Add HomeServlet to handle requests to the main page
IT26100781 committed 2 weeks ago
4318f63
Add StaffLogoutServlet to handle admin staff logout functionality
IT26100781 committed 2 weeks ago
1fa5173
Add StaffAuthFilter to enforce staff authorization for admin routes
IT26100781 committed 2 weeks ago
e2a908d
Add AdminDashboardServlet for admin dashboard
IT26100781 committed 2 weeks ago
e1a593f
Add StaffSetupServlet to handle first admin setup process
IT26100781 committed 2 weeks ago
dbd6ae9
Create core JSP views for public and admin interfaces, including dashboard, error, and main pages.
IT26100781 committed 2 weeks ago
5ebad42
Commits on Oct 27, 2026
Remove file-based methods from Staff and User classes
IT26100781 committed 2 weeks ago
f5b8da1
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Issues
Pull requests
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Security and quality
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Settings
Commits
main
All users
All time
Commits on Nov 10, 2026
Merge pull request #13 from IT26100781/Ui-improvement
IT26100781 authored yesterday
Verified
755d1ce
Improve reviews UI
IT26100781 committed yesterday
fec1b0b
Merge remote-tracking branch 'origin/main' into Ui-improvement
IT26100781 committed yesterday
7e001df
Merge pull request #12 from IT26100781/reservation
IT26100781 authored yesterday
Verified
59680f0
fix(reservation): validate arrival and checkout dates before allowing check-in
it26100826 committed yesterday
6112215
feat(reservation): implement STI inheritance (Online/Desk), booking servlets, dashboard metrics, and schema fix
it26100826 committed yesterday
6bd37bd
merge main, accept main JSPs/schema, remove junk log file
it26100826 committed yesterday
29f8122
Merge pull request #11 from IT26100781/payment
IT26100781 authored yesterday
Verified
ab76f6d
Delete src/main/java/com/hotel/.DS_Store
IT26101177 authored yesterday
Verified
1bfe5ed
Merge pull request #10 from IT26100781/reviews
IT26100781 authored yesterday
Verified
e173078
add jsps and dao
it26100826 committed yesterday
5a0ef88
jsp fix
IT26100939 committed yesterday
e4829b5
bug fixes
IT26100939 committed yesterday
d8b464e
added sql
IT26100939 committed yesterday
8475bad
merge origin main to payment
IT26101177 committed yesterday
52a00a8
Merge remote-tracking branch 'origin/main' into payment
IT26101177 committed yesterday
7f55633
bug fixes and code formatting
IT26100939 committed yesterday
e981d36
Add booking and payment JSP views
IT26100781 committed yesterday
b11de1a
update the codes
IT26101177 committed yesterday
12fb860
Commits on Nov 9, 2026
Add write.jsp to project
IT26100939 committed 2 days ago
eed6a9d
Add moderate.jsp to project
IT26100939 committed 2 days ago
617e466
Add Review.sql to project
IT26100939 committed 2 days ago
fa55835
Merge pull request #8 from IT26100781/reservation-payment-ui
IT26100781 authored 2 days ago
Verified
6a63f25
Add booking and payment JSP views
IT26100781 committed 2 days ago
cdc795c
Merge pull request #7 from IT26100781/admin
IT26100781 authored 2 days ago
Verified
c760bca
Commits on Nov 8, 2026
Add ReviewModerateServlet to project
IT26100939 committed 3 days ago
fd1adc5
Add ReviewDeleteServlet to project
IT26100939 committed 3 days ago
49138ed
Add ReviewSaveServlet to project
IT26100939 committed 3 days ago
5d046c4
Add ReviewFormServlet to project
IT26100939 committed 3 days ago
9df9ac0
Add admin home page content editor
IT26100781 committed 3 days ago
3ae9211
Add ReviewListServlet to project
IT26100939 committed 3 days ago
602fba3
Add Review service to project
IT26100939 committed 3 days ago
a8fb2a8
Commits on Nov 7, 2026
Add ReviewDAO to project
IT26100939 committed 4 days ago
8714434
Add Verified Review to project
IT26100939 committed 4 days ago
1d68637
Add Public Review to project
IT26100939 committed 4 days ago
ea10f19
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